



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Supplementary Examinations, July-2025

DIGITAL ELECTRONICS

(COMMON TO CSE, IT, CSE (DS))

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) Convert a 2 input binary number into a gray code. [1M] L2
- b) State De Morgan's Theorem. [1M] L1
- c) Implement the given Boolean function in NAND-NAND Logic $Y=A'+BC$. [1M] L3
- d) Minimize the expression using K Map $f(A,B,C) = \sum(1,2,4,5,7)$. [1M] L3
- e) Design a OR gate using suitable Multiplexer. [1M] L3
- f) Draw the structure of two bit binary multiplier. [1M] L2
- g) Differentiate Latch and a Flip flop. [1M] L2
- h) What are sequential Circuits? Give examples. [1M] L2
- i) What are Hazards? [1M] L1
- j) What are the different types of Random Access Memories? [1M] L1

PART- B

(5 X 10 =50 Marks)

2. a) convert the following hexa-decimal --> decimal --> octal

(1) (ABCD)₁₆ to ()₈

(2) (4056)₈ to ()₂

[5M] L3

b) Write Short Notes on Binary Codes.

[5M] L3

OR

3. a) Convert the following expression to POS form $F(A,B,C) = (A'+B')ABC$. [5M] L3

b) Simplify $f(A,B,C) = A'C' + A'B'C' + ABC + ABC'$. [5M] L3

4. a) Minimize using K Map $f = AB + BC + AC' + AD + AB'C + ABC$. [5M] L3

b) Minimize using K Map $f(A,B,C,D) = \sum(1,3,5,8,9,11,15) + d(2,13)$. [5M] L3

OR

5. a) Minimize using K Map $F(A,B,C,D) = \pi(1,2,3,8,9,10,11,14) + d(7,15)$ $A'C'(1+B') + AB(C+C')$ [5M] L3

b) Minimize using K Map $f(A,B,C,D) = \sum(2,4,6,8) + d(10,11,12,13,14,15)$ $A'C' + AB$ [5M] L3

6. a) Explain the Binary Full adder with Suitable diagrams and Boolean Equations. [5M] L3
b) Design a 2 Bit Magnitude Comparator. [5M] L3

OR

7. a) Design a 4 X 1 Multiplexer. Explain with its truth table. [5M] L3
b) Design a 3 to 8 Decoder using basic gates. [5M] L3

8. Explain about JK-Flip flop with its characteristic equation. What is Race around condition and how it is eliminated. [10M] L3

OR

9. Design a 3 Bit Asynchronous Up Counter using T Flip flop. [10M] L3

10. Enumerate Program Array Logic and Program Logic Array. [10M] L1

OR

11. Explain with an example Asynchronous sequential circuit design. [10M] L2

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY**(AN AUTONOMOUS INSTITUTE)****Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029****B.Tech II Year I Semester Regular Examinations, January-2025****DIGITAL ELECTRONICS****(COMPUTER SCIENCE AND ENGINEERING)****Time: 3 hours****Max. Marks: 60****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A**(10 Marks)**

1. a) Using Binary multiplication rules Multiply $(1101.1)_2$ with $(101)_2$ [1M] L2
- b) $(10110110110)_2$ Find Gray Code equivalent? [1M] L3
- c) Simplify the Boolean expression to minimum number of literals: $(A+B)' (A'+B')$ [1M] L1
- d) What is the use of don't care combinations? [1M] L2
- e) What is priority encoder? [1M] L1
- f) What is the function of a multiplexer's select input? [1M] L2
- g) What is a flip-flop? Write down the characteristic equation of T flip-flop. [1M] L1
- h) Discuss the difference between Combinational and Sequential circuits. [1M] L2
- i) State the types of ROM? [1M] L1
- j) Write a short notes on State Assignment. [1M] L2

PART- B**(5 X 10 =50 Marks)**

2. a) Express the following numbers in decimal: $(10110.0101)_2$, $(16.5)_{16}$, $(26.24)_8$ [5M] L1
- b) Evaluate the 7-bit hamming code to the given 4bit data word 1011 that corrects and detects single bit error. [5M] L3

OR

3. a) Give the truth table for the Boolean Expression $(x'+y)'$ [5M] L2
- b) What are universal gates and implement the basic gates using universal gates. [5M] L3
4. Simplify the following function using K-Map. Implement the simplified function using Nand gates
 $F = \sum m(0,1,3,6,9,10,12,13,16,17,20,30,31) + \sum d(7,14,21)$ [10M] L3

OR

5. a) Find the minimal Boolean expression of $F = \sum m(0,1,2,3,4,5,6,7,8,10,11,12,13,14,15)$ [5M] L2
- b) Define Pair, Quad and Octet in K-Maps and give examples [5M] L1

6. a) Implement a Full adder using two half adders and one OR gate. [5M] L2
b) Assess the implications of using a 4-to-2 encoder in a digital system [5M] L3

OR

7. a) Design a 4-Bit Binary to Gray code Converter using logic gates? [7M] L4
b) What is a 1-bit magnitude comparator? Draw its logic diagram. [3M] L1
8. a) What is an excitation table of a flip-flop and draw the transition table for the flip-flops
T-F/F, S-R-F/F? [5M] L1
b) Convert SR Flip-Flop into JK Flip-Flop. [5M] L2

OR

9. a) Design a Johnson counter and explain its working with the help of timing diagram [5M] L3
b) Design a 3-bit synchronous down counter using T- flip-flops. [5M] L2

10. a) Explain different types of memory. [5M] L1
b) Design a Random Access memory having 8 K Bytes. Identify how many address lines are needed.

[5M] L3

OR

11. Explain in detail about PLD's.

[10M] L2

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PART- A

(10 Marks)

1. a) What do you mean by non-weighted codes? Give examples. [1M] L1
- b) Subtract the given numbers using 1's complement?
i.) 110110 – 100111 ii.) 0100 – 00011 [1M] L3
- c) Distinguish between canonical and standard forms by giving an example. [1M] L1
- d) Map the given expression $f(A,B,C)=\sum m(0,2,3,4,5,6)$ [1M] L1
- e) Draw the logic circuit of a Half adder and give its truth table. [1M] L1
- f) Convert the following function $f(a,b,c)=a'b+b'c'$ into SSOP [1M] L4
- g) Write the Excitation table of a J-K flip-flop? [1M] L1
- h) Discuss the difference between synchronous and asynchronous sequential circuits. [1M] L2
- i) Write a short notes on State table of sequential circuits [1M] L1
- j) List the basic types of Programmable Logic Devices? [1M] L1

PART- B

(5 X 10 = 50 Marks)

2. The message below was coded in the Hamming code with even parity and transmitted through a noisy channel. Decode the message assuming that a single bit error has occurred in each code word
1011001 [10M] L3

OR

3. a) State and prove DeMorgan's Theorems [5M] L3
- b) Draw the logic symbols and truth tables of XOR and XNOR gates [5M] L2
4. Simplify the following 5 variable SSOP function
 $F(A,B,C,D,E)=\sum m(0,5,6,8,9,10,11,16,20,24,25,26,27)$ using K map? [10M] L3

OR

5. a) Simplify the following Boolean expressions to a minimum number of literals
i) $ABC+A'B+ABC'$ (ii) $xy + x(wz+wz')$ (iii) $Y=(A'+B')(A+C')+B'(B+C)$ [5M] L2

b) Assess how the knowledge of Boolean function minimization through K-maps can impact the efficiency of digital circuit design [5M] L3

6. With a neat diagram explain in detail about Decimal Adder. [10M] L3

OR

7. a) Implement the following Boolean function using suitable Multiplexer $F(A,B,C) = \sum m(1,3,5,6,7)$ [5M] L4

b) Implement AND, OR, NOR by using NAND gates only. [5M] L1

8. a) What is the characteristic equation of a S-R flip-flop based on its characteristic table and how is it derived? [5M] L2

b) Convert SR Flip-Flop into D Flip-Flop [5M] L2

OR

9. Explain Race Around condition & how it is eliminated in Master Slave JK Flip-flop. [10M] L3

10. Design a PLA for the following equation. $F = a'bc + b'c + ab$ [10M] L5

OR

11. Explain Various Memory Decoding Techniques. [10M] L2

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